



9.3.5 Discussion ¶

To summarize this section:

- The Power Method finds the eigenvector associated with the largest eigenvalue (in magnitude). It requires a matrix-vector multiplication for each iteration, thus costing approximately $2m^2$ flops per iteration if A is a dense $m \times m$ matrix. The convergence is linear.
- The Inverse Power Method finds the eigenvector associated with the smallest eigenvalue (in magnitude). It requires the solution of a linear system for each iteration. By performance an LU factorization with partial pivoting, the investment of an initial $O(m^3)$ expense then reduces the cost per iteration to approximately $2m^2$ flops. if A is a dense $m \times m$ matrix. The convergence is linear.
- The Rayleigh Quotient Iteration finds an eigenvector, but with which eigenvalue it is associated is not clear from the start. It requires the solution of a linear system for each iteration. If computed via an LU factorization with partial pivoting, the cost per iteration is $O(m^3)$ per iteration, if A is a dense $m \times m$ matrix. The convergence is quadratic if A is not Hermitian, and cubic if it is.

The cost of these methods is greatly reduced if the matrix is sparse, in which case each iteration may require as little as $O(m)$ per iteration.